



Physiological breeding for yield improvement in soybean: solar radiation interception-conversion, and harvest index

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Abstract

Key message Efficiency of light interception, Radiation use efficiency and harvest index can be used as targets to improve grain yield potential in soybean.

Abstract Grain yield (GY) production can be expressed as the result of three main efficiencies: light interception (Ei), radiation use (RUE), and harvest index (HI). Although dissecting GY through these three efficiencies is not entirely new, there is a lack of knowledge about the phenotypic variation, the genetic architecture, and the relative contribution of these three efficiencies on GY in soybean. This knowledge gap coupled with laborious phenotyping prevents the active consideration of these efficiencies into breeding programs. This study aims to reveal the phenotypic variation, heritability, genetic relationships, genetic architecture, and genomic prediction for Ei, RUE, and HI in soybean. We evaluated a maturity control panel of 383 Recombinant Inbred Lines (RILs) selected from the soybean nested association mapping (SoyNAM) population. Dry matter ground measured along with canopy coverage (CC) from UAS imagery were collected in three environments. Light interception was modeled through a logistic curve using CC as a proxy. The total above-ground biomass collected during the growing season and its respective cumulative light intercepted were used to derive RUE through linear models fitting. Additive-genetic correlations, genome-wide association (GWA) and whole-genome regressions (WGR) were performed to evaluate the relationship between traits, their association with genomic regions, and the feasibility of predicting these efficiencies with genomic information. Correlation analyses considered three groups: the entire data set, and the high- and low-yielding RILs to determine association as a function of the GY. Our results revealed moderate to high phenotypic variation for Ei, RUE, and HI with ranges of 8.5%, 1.1 g MJ⁻¹, and 0.2, respectively. Additive-genetic correlation revealed a strong relationship of GY with HI and moderate with RUE and Ei when whole data set was considered, but negligible contribution of HI on GY when just the top 100 was analyzed. The GWA analyses showed that Ei is associated with three SNPs; two of them located on chromosome 7 and one on chromosome 11 with no previous quantitative trait loci (QTLs) reported for these regions. RUE is associated with four SNPs on chromosomes 1, 7, 11, and 18. Some of these QTLs are novel, while others are previously documented for plant architecture and chlorophyll content. Two SNPs positioned on chromosome 13 and 15 with previous QTLs reported for plant height and seed set, weight and abortion were associated with HI. WGR showed high predictive ability for Ei, RUE, and HI with maximum correlation ranging between 0.75 and 0.80. Future improvements in GY can be expected through strategies prioritizing Ei for short-term results when using high yielding germplasm and RUE for medium- and long-term outcomes. This work is a pioneer attempt to integrate traditional physiological traits into the breeding process in the context of physiological breeding.

Introduction

Yield in cultivated crops can be expressed as a result of three main physiological processes: solar radiation interception by the canopy, solar radiation conversion into biomass, and harvest index (Monteith 1972, 1977). The proportion of solar radiation intercepted also referred as the Ei is the result of genetic and agronomic factors. Canopy architecture including leaf angle, leaf size, and leaf area index (LAI) (Monteith

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1969; Chen et al. 1994; Bai et al. 2016; Chavarria et al. 2017), as well as pigment composition (Vogelmann y Evans 2002; Taiz et al. 2014), and stay-green (Thomas y Ougham 2014; Hammer et al. 2014; Shi et al. 2016; Christopher et al. 2016) are the main genetic properties influencing this efficiency. From the agronomic perspective, distance between plants, distance between rows (Andrade et al. 2002; Edwards et al. 2005; Shibles and Weber 1966; Westgate et al. 1997), and nutrients and water supply (De Costa y Shanmugathan 2002; Sandaña et al. 2012) are the factors primarily accounting for *es* in light interception patterns.

Solar radiation conversion or RUE is a function of the type of metabolism involved in the reduction and consumption of carbon through the processes of photosynthesis and respiration (Beadle y Long 1985; Gosse et al. 1986; Loomis y Amthor 1999; Sinclair y Muchow 1999; Reynolds et al. 2000; Liu et al. 2015). Environmental offer, particularly light available and CO₂ concentration also influences RUE since their influence in carbon fixation and respiration (Sinclair y Horie 1989; Stockle y Kiniry 1990; Sinclair et al. 1992; Jamieson et al. 1995; Purcell et al. 2002; Dermody et al. 2008; Asif et al. 2010; Quanqi et al. 2012). In soybean, doubling the RUE is still possible, since the theoretical maximum was estimated to be 9.4% and current values range from 2.3 to 4.3% (Beadle et al. 1987; Long et al. 2006; Zhu et al. 2008, 2010).

Finally, HI measures the reproductive effort and that is closely related to the sink-source relationship (Donald y Hamblin 1976; Hay 1995). It is considered that most crop yield gains achieved during the last six decades exploited direct or indirect improvements in HI or total biomass produced (Donald y Hamblin 1976; Hay 1995; Evans et al. 1999; Sadras y Lawson 2011). The role of HI in soybean GY is inconsistent; however, negative changes have not been reported in HI for increasing GY (Rowntree et al. 2014; Suhre et al. 2014; Balboa et al. 2018). Although the influence of HI in yield improvement is widely accepted, future GY gains cannot rely solely on this parameter. Rather they must involve the enhancement of the other efficiencies, particularly RUE (Loomis y Amthor 1999; Lindquist et al. 2005; Foulkes y Reynolds 2014). It has been hypothesized the improvement on HI has a maximum theoretical estimated to be 0.60 (Evans et al. 1980; Foulkes et al. 2011) offering an absolute gap to improve of at least ~ 10% in most crops (Foulkes et al. 2011).

Soybean perspectives in the medium- and long-term are promising considering future demand for protein and oil for the rising human, animal, and fuel consumption (Jaggard et al. 2010; Alexandratos y Bruinsma 2012; Ray et al. 2013; Bao et al. 2015). Likewise, new projected demand in Africa that, is expected to turn soybean into a dominant crop given its nutritional profile, the unsatisfied demand (6.8 Mt imported for Sub-Saharan Africa), and the added

oil production, suggest an auspicious upcoming for this crop (Khojely et al. 2018; Foyer et al. 2019; Gbegbelegbe et al. 2019). However, this favorable scenario contrasts with the moderate grain yield gains obtained during the last decades calculated in 22–27 kg ha⁻¹ yr⁻¹ (USDA-ERS 2011; Fox et al. 2013; Koester 2014). This panorama where new demand exceeds the rate of increase in production represents an opportunity to explore new complementary strategies for yield improvement.

Despite the known theoretical framework about the influence of Ei, RUE, and HI on final yield (Reynolds et al. 2000, 2012a; Foulkes y Reynolds 2014), the phenotypic variation and the genetic architecture for these efficiencies in soybean is not well established limiting their active inclusion into breeding programs. Likewise, the relative contribution of each efficiency to the final grain yield and the genetic relationship among them are not quantified with potential implications in breeding processes including prioritization of traits, indirect selection or genetic drag. In this study, we aimed to overcome this knowledge gap quantifying the phenotypic variation for Ei, RUE, and HI in soybean through a set of field experiments involving 383 recombinant inbred lines (RILs) from the SoyNAM collection. In addition, we revealed the genetic architecture for Ei, RUE, and HI and their additive-genetic correlations with GY. Finally, the feasibility to implement genomic prediction-selection as a strategy to overcome the phenotyping bottleneck was also explored.

Materials and methods

Plant material and field design

A set of 383 RIL coming from 32 families of the SoyNAM collection were evaluated in three environments during the years 2017 and 2018. These 32 families were also originally classified into three classes: high yielding (HY), high yielding under drought conditions (HYD), and diverse ancestry (DA). Information about the parents, complete description of crosses, and extra information is available in <https://www.soybase.org/SoyNAM/> and Supplementary Table 1 and 2, respectively. The RILs' selection criteria aligned with creating a panel with variation for yield (3088–4396 kg ha⁻¹) while constraining maturity to ± 2 days (Supplementary Fig. 1). Data were collected in two different location: the Purdue University Agronomy Center for Research and Education-ACRE (40° 28' 20.5" N 86° 59' 32.3 " W) at West Lafayette, IN and Romney, IN (40° 14' 59.1" N 86° 52' 49.4" W)-RMN. In this study, the combination of location $\times$ year is considered as environment: ACRE_2017, ACRE_2018, and RMN_2018. Field design corresponded to an alpha lattice incomplete block design, with two complete

replications and 32 incomplete blocks per replication. Six rows plots (0.76 m × 3.35 m) planted with a target population of 35 seed m⁻² constituted the experimental unit. Plots with non-uniform emergence were discarded reducing the number of RILs to 322 for ACRE_2017 and 381 for RMN_2018. Experiment were planted on 5/30/2017, 5/22/2018/, and 5/17/2018 for ACRE_2017, ACRE_2018, and RMN_2018, respectively. No inoculation treatment to the seeds was implemented. Soil types for ACRE included Chalmers silty clay loam (Typic Endoaquolls) and Raub-Brenton complex (Aquic Argiudolls), while RMN corresponded to Drummer soils (Typic Endoaquolls) (NRCS 2018). Adequate nutritional status during the growing season was ensured through the crop management, along with the high natural soil fertility confirmed through the soil analysis (Supplementary Table 3). Mean precipitation reached 132, 130, and 91 mm/month for ACRE_2017, ACRE_2018, and RMN_2018, respectively (iClimate 2019), whereas adequate water availability was confirmed through water balance for each location (Supplementary Tables 4–6).

Traits measurement

Solar radiation interception measured as the fraction between the light on the bottom of the canopy compared with the top was estimated through the canopy coverage (CC) obtained from unmanned aircraft system (UAS) imagery. canopy coverage as a proxy for light interception (1:1 relationship) has been previously reported and validated in soybean for Kawasaki et al. (2016) and Purcell (2000), we also checked this approach measuring light interception in 72 plots using a line quantum sensor LI-191R (LICOR Inc, Lincoln, NE). Average per plot of three reads below canopy were collected, while two reads above canopy were taken. The validation protocol was implemented between 11:00 and 2:00 pm, the same day and immediately before and after the UAS' flight. The plot extraction and canopy quantification were accomplished using a multilayer approach where individual pictures for each plot rather than average orthomosaic were analyzed (Hearst 2019) through the software Progeny® (Progeny Drone Inc., West Lafayette, IN). This approach uses all the pictures collected for each plot gaining accuracy and enabling to work with distribution for the canopy coverage rather than a single value. Flights were carried out between 11:00 am and 2:00 pm in clear sky days to minimize solar angle and clouds effect during the plot extraction and canopy quantification. A fixed-wing UAS type eBee equipped with an S.O.D.A red–green–blue (RGB) camera (senseFly Parrot Group, Switzerland) was flown after every ~ 12 days. Six to eight flights per growing season were considered to model the canopy growth by implementing a logistic regression, Eq. (1), using the R software (R Core team 2019) package “growthrates” (Petzoldt 2020).

$$y = \frac{k * y_0}{y_0 + (k - y_0) * \exp^{-\mu \max * time}} \quad (1)$$

where y is canopy coverage, y_0 is the minimum canopy coverage value measured, k corresponds to the maximum canopy value or load capacity, $\mu \max$ is the maximum relative growth rate, and time indicates days after planting. In this case, and considering the 1:1 relationship between canopy coverage and light intercepted already described, the logistic canopy coverage models are equal to the light intercepted models.

Logistic models allowed us to transform a “discrete” (after every ~ 12 days) into a “continuous” (daily) variable for light interception, which aligns with the daily solar radiation coming from the weather station. Thus, the amount of light intercepted per plot was determined as the product of multiplying the daily fraction of canopy coverage (light intercepted), derived from the fitted logistic regressions, by the daily photosynthetic active radiation (PAR) available, 48.7% of total solar radiation (Zhu et al. 2008). Solar radiation data from weather stations located on site (ACRE) and less than one mile from the experimental field (RMN) were used (iClimate 2019). Finally, the efficiency of light interception was calculated as the ratio between the cumulative PAR intercepted and the cumulative PAR available during the growing season.

RUE was calculated as the slope of the linear regression between dry matter and cumulative PAR intercepted during the growing season. Dry matter accumulation was determined by sampling all the soybean plants present in a linear section of 0.56 m in a row with perfect competition; it is a row flanked in all the four cardinal points with the same cultivar. The fresh biomass collected from each sampling site was dried at 80 °C using a dry air system until a constant weight was achieved. Constant weight was verified through the none difference between two successive daily sampling weights for the same set of samples. The final dry weight was registered using a precision balance NV1201 (Ohaus, Parsippany, NJ). Three full biomass sampling: ~ 38, 58, and 84 days after planting (DAP) were considered for both environments in 2018, while just one full sampling at 91 DAP was carried out in 2017. Then, RUE for ACRE_2017 corresponds to the simple ratio between biomass and cumulative solar radiation at 91 DAP. To avoid potential differences in dry matter associated with the number of plants sampled each time, biomass data per sampling were adjusted through a simple linear model involving RIL, environment, and replication as variables and number of plants as covariate.

Apparent harvest index defined as the ratio between the dry matter in the seeds and the total dry matter above-ground (Hay 1995) was calculated at R8 growing stage (Fehr and Caviness 1977) in a linear sample of 0.56 m of row. This is called apparent harvest index rather than harvest index since

root dry matter was not obtained. The above-ground biomass harvested was dried following the procedure already mentioned to obtain dry matter. Likewise, seed moisture was estimated using the NIR equipment Perten DA 7200 (Perten Instruments Inc., Springfield, IL) and subsequently adjusted to zero moisture.

Grain yield, in turn, was collected from two central rows for each field plot using a combine harvester. Moisture in seed grain was determined using the NIR equipment Perten DA 7200. Contrary to HI, in this case the moisture was adjusted to 13% in all the cases and the grain yield extrapolated to hectare. Distribution for all the traits including the p value for the Kolmogorov–Smirnov test for normality is presented in the Supplementary Fig. 2.

Statistical model and data analyses

A mixed model through the R package ‘*lme4*’ (Bates et al. 2015) was implemented to analyze collected field data. The statistical model included environment (combination of year $\times$ location), replication, block, and RIL as sources of variation. In addition, a spatial covariate $f(x)$ accounting for field heterogeneity was considered. We computed this covariate $f(x)$ as the average phenotypic value from the four plots immediately adjacent to each of our sampling plot (Lado et al. 2013) using the function *NNsrc* from the R “*NAM*” package (Xavier et al. 2015). The model implemented corresponds to

$$Y_{ijkl} = \mu + f(x) + \alpha_i + \beta_{j(i)} + (\beta\gamma)_{jk} + \delta_l + e_{ijkl} \quad (2)$$

where Y is the vector of phenotypes measured in the i th environment, into the j th replication in the k th block for the l th RIL, μ is the intercept, $f(x)$ controls the spatial heterogeneity within replications, α accounts for the effect of the environment, β controls the effect of replication nested into the environment, $\beta\gamma$ corresponds to the interaction replication $\times$ block, δ accounts for the genetic effect, and e controls the error. In this model, the spatial covariate and the RILs were treated as fixed effects, while the other sources of variation were considered as random where any random effect, $\sim N(0, \sigma^2_r)$, and $e \sim MVN(0, R)$.

Genomic information

This study used already existing genomic information; founder parents from the SoyNAM collection were both genome sequenced and genotyped with the soySNP50K BeadChip (Song et al. 2013). Similarly, the entire SoyNAM collection was also genotyped with a SoyNAM6K BeadChip SNP with 5305 SNPs specially design (Song et al. 2017b). The genomic information from both SNP BeadChip

was combined to project the segregating markers from SoySNP50K onto the SoyNAM RILs using as a framework the mapped SoyNAM6K markers using the software *finhap f90* (VanRaden et al. 2015). Positions in base pairs (bp) from “Williams 82” reference genome (Wm82.a2.v1) were used. As a result of this process, 42,509 SNP markers were detected. Considering our medium size RILs’ panel, a quality control using a minor allele frequency less than 15% (MAF < 0.15) was implemented (Jarquín et al. 2014; Xavier et al. 2016) reducing our data set to 23,119 SNPs. This level of MAF ensures comparisons where the rare allele counts with at least 57 accessions. The original allele frequency plot, as well as the representation of each RIL in the principal component plot, are presented in the Supplementary Fig. 3.

Association analysis

Genome-wide association analyses were performed following a whole-genome regression approach where all the SNPs (23,119) were incorporated simultaneously to the model preventing potential problems linked with multiple testing (Xavier et al. 2018, 2020). The implemented model is as follow

$$y = \mu + Xb + e \quad (3)$$

where y is the response variable, μ is the intercept, X is the genotypic matrix, b is the regression coefficient or effect of an allele substitution, with d probability of being included into the model, and e is the residual term. In our case, best linear unbiased estimators (BLUEs) from the model (1) were considered as the response variable (y), rather than observed phenotypes (Möhrling y Piepho 2009; Rosyara et al. 2016). The association between markers and traits was evaluated using as threshold a p value of 0.03. Since we implemented WGR approach, multiple testing correction was not needed. Whole-genome regressions were fitted using the R package Bayesian Whole-Genome Regression “bWGR” using the function “*BayesDPI*” (Xavier et al. 2020) with variable selection based on Metropolis–Hastings and polygenic term solved via Bayesian algorithm of reproducing kernel Hilbert Spaces (De Los Campos et al. 2010; Xavier et al. 2018, 2020). Circular Manhattan plots using the R package “*CM-plot*” (LiLin-Yin 2020) were used to display the results. Potential candidates genes were explored using the window of linkage disequilibrium (LD) reported for each chromosome (Wen et al. 2015).

Variance components and genetic correlations

Variance components were estimated from model (1) using the R package “*lme4*” (Bates et al. 2015). However, in this

case, RIL factor was treated as a random effect. Once variance components were estimated, the broad-sense heritability on entry mean basis is calculated following Eq. (4)

$$H = \frac{V_g}{V_g + \frac{V_e}{r}} \quad (4)$$

where H corresponds to broad-sense heritability on an entry mean basis, V_g is genetic variance, V_e corresponds to variance of error, and r is the number of replications.

In addition, a multivariate mixed model (MMM) for standardized values using a Markov Chain Monte Carlo (MCMC) approach was implemented using the R package “MCMCglmm” (Hadfield 2015) following the model presented in Eq. (2) with RIL as a random effect. In this model, we accounted for the additive-genetic effects through a relationship matrix calculated from the data set of SNPs. The function GRM from the “NAM” package was used (Xavier et al. 2015). Genetic correlations among traits and narrow-sense heritability (h^2) were calculated from the variance–covariance matrices produced after 50,000 iterations with 5000 discards (burn in). Trace and multicollinearity among chains were inspected as quality checks for the model. Narrow-sense heritability is calculated following the formula shown in Eq. (5)

$$h^2 = \frac{V_a}{V_a + V_e} \quad (5)$$

where h^2 corresponds to narrow-sense heritability, V_a is additive-genetic variance, and V_e corresponds to the variance of error.

To establish the relationship among efficiencies for “high” and “low” yielding lines, we also performed the same multitrait analysis for two extra data sets made up of the top 100 and bottom 100 RILs selected by average grain yield over environments.

Results

Phenotypic variation, heritability, and genetic correlations

Ei was found to have a mean value for the three locations of 50.7%, with minimum and maximum of 45.3% and 53.8%, respectively (Fig. 1A). Families with HYD parents showed the highest average Ei with 51.4%, followed by DA and HY both with 50.7%. ACRE_2017 reached the highest mean interception with 56.3%, while its range varied between 52.2 and 59.3%. Contrarily, RMN_2018 experienced the lowest light intercepted with 44.6% as its mean value. The minimum and maximum for this environment were 39.1% and

50.2%, respectively. ACRE_2018, in turn, presented a mean interception value of 52.8% with minimum and maximum values of 48.1% and 57.0%, respectively. These results suggest there is still genetic variation for Ei that can be exploited through plant breeding which should contribute to higher GY considering the positive relationship between Ei and GY.

Consolidated RUE for the three environments ranged between 1.87 and 3.01 g MJ⁻¹, with a mean value of 2.40 g MJ⁻¹ PAR (Fig. 1B). HY families presented the highest mean RUE values with 2.43 g MJ⁻¹ followed by DA families with 2.40 g MJ⁻¹, while HYD reported the lowest value with 2.34 g MJ⁻¹. Environments evaluated during 2018 had the highest mean RUE with 2.46 g MJ⁻¹ PAR and 2.59 g MJ⁻¹ PAR for ACRE_2018 and RMN_2018, respectively. ACRE_2018 ranged between 1.91 g MJ⁻¹ PAR and 3.05 g MJ⁻¹ PAR, while minimum and maximum values for RMN_2018 were found in the order of 1.98 g MJ⁻¹ PAR and 3.17 g MJ⁻¹ PAR. ACRE_2017 presented the lowest conversion efficiency with a minimum, mean, and maximum value of 1.48, 2.04, and 2.54 g MJ⁻¹ PAR, respectively. These results indicate a moderate to high environmental influence on RUE, but also suggest the improvement of this trait as a viable strategy to increase GY in soybean since HY progenies frequently showed higher RUE.

Apparent HI showed an overall mean of 0.39 with minimum of 0.28 and maximum values of 0.42 (Fig. 1C). HY families presented the highest HI with a mean value of 0.40, while DA and HYD families reported average values of 0.39 and 0.37, respectively. At the level of environment, the highest mean HI was obtained in ACRE_2017 with 0.44 followed by RMN_2018 and ACRE_2018 with 0.38 and 0.36, respectively. Minimum and maximum values followed the same trend regard environments with 0.33, 0.28, and 0.24 for minimum and 0.53, 0.49, and 0.48 for maximum in ACRE_2017, RMN_2018, and ACRE_2018, respectively. Our findings highlight the transcendental role that HI improvement has played in the way of higher GY in soybean. However, the fact that the overall mean and maximum values are relatively close each other suggest a less chance of capitalizing in better soybean material through HI.

GY was found to have an overall mean of 3665 kg ha⁻¹, while its minimum and maximum reached 2339 and 4942 kg ha⁻¹, respectively. Regarding the three classes of families considered, HY showed the highest average GY value with 3846 kg ha⁻¹ contrasting with HYD with 3071 kg ha⁻¹. Mean value for DA families, in turn, reached 3577 kg ha⁻¹. ACRE_2017 was the location with the lowest mean GY with 3178 kg ha⁻¹, with minimum and maximum of 1673 and 4606 kg ha⁻¹. The environment in 2018 presented similar performance with mean GY of 3797 and 3886 kg ha⁻¹ for ACRE_2018 and RMN_2018, respectively. Minimum values, in turn, corresponded to 2474 and 2505 kg ha⁻¹, while maximum reached 5160 and 5342 for, in

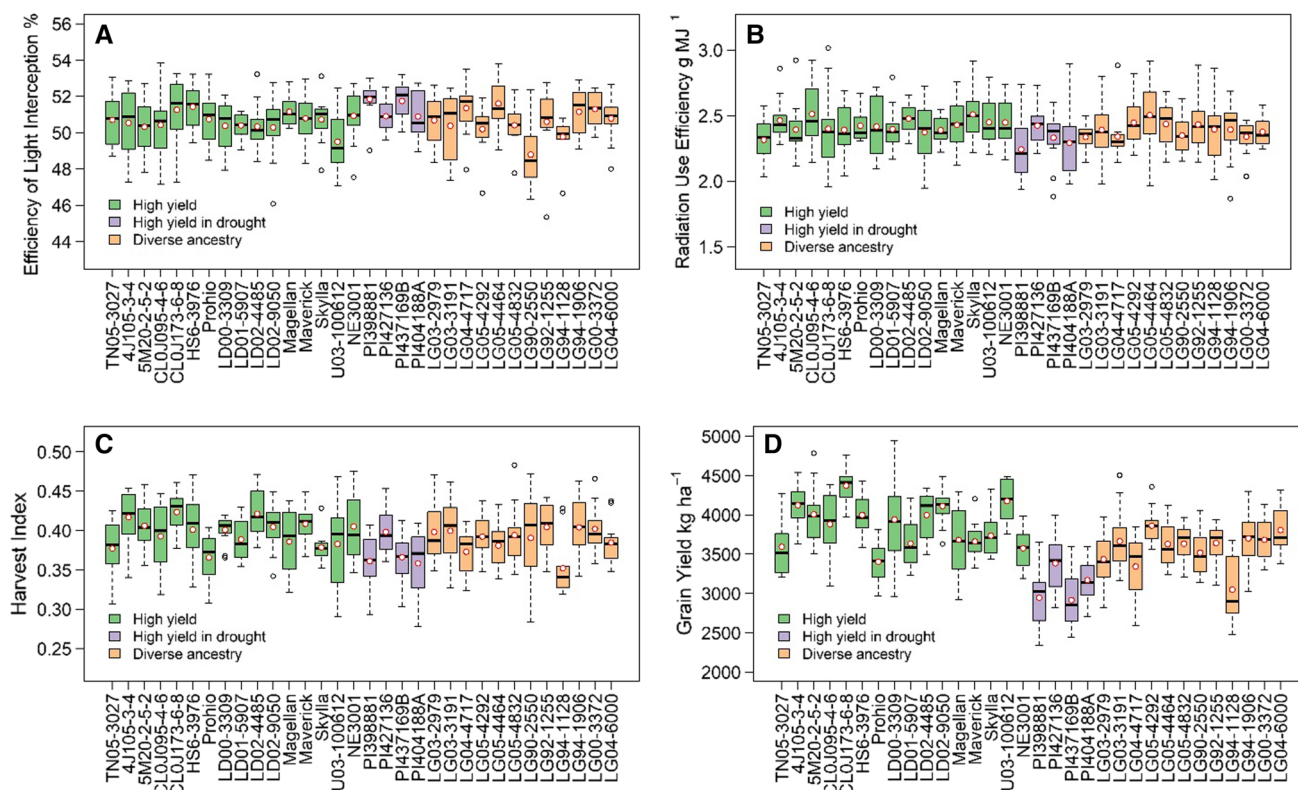


Fig. 1 Phenotypic variation for light interception efficiency **A**, radiation use efficiency **B**, harvest index **C** and grain yield **D** grouped by family in a soybean maturity-controlled panel. Three hundred and eighty-three Recombinant Inbred Lines (RIL), 12 RILs per family, and three environments. Average over environments. Colors repre-

sent the type of population assigned to the parent when the SoyNAM panel was developed. Red circles denote the mean value, horizontal lines in the box indicate the median, dashed lines represent the minimum and maximum values, and empty circles correspond to outliers

both cases, ACRE_2018 and RMN_2018. Significant statistical differences ($p < 0.01$) among RILs, families, and classes (HY, HYD, DA) were found for all three efficiencies and grain yield. These results show the genetic background plays a fundamental role to obtain high yielding soybeans. Likewise, they also suggest a potential contribution of diverse ancestry material in high yielding commercial soybeans.

Heritability on entry mean basis was moderate to high (Table 1). Grain yield was found to have the highest value with 0.82, contrasting with radiation use efficiency, which reached a value of 0.34. Ei and HI presented values of 0.62 and 0.58, respectively. Additive-genetic effects accounted for 24.0–65.7% of the phenotypic variation when the whole data set was considered. RUE was the trait with less variance proportion explained followed by Ei and HI with 41.4 and 48.0%. GY showed the highest narrow-sense heritability (Table 1). When top (100) and bottom (100) data sets were analyzed, changes in narrow-sense heritability were obtained. While GY remained almost constant with 0.68 and 0.62 for top and bottom, Ei increased from 0.42 to 0.56 when top and bottom data set were considered. RUE, in turn, changed from 0.40 in the top 100 RIL to 0.25 in the bottom

Table 1 Broad-sense heritability on plot basis (H) and narrow (h^2) sense heritability for light interception efficiency (Ei), radiation use efficiency (RUE), harvest index (HI), and grain yield (GY) for three data sets: All, Top 100, and Bottom 100 in a maturity-controlled panel of soybean.

Trait	H-All	h^2 -All	h^2 -top 100	h^2 -bottom 100
Ei	0.62	0.41	0.42	0.56
RUE	0.34	0.24	0.40	0.25
HI	0.58	0.48	0.06	0.62
GY	0.82	0.66	0.68	0.62

Data set “All” includes three hundred and eight-three recombinant inbred lines RIL. Three environments in all the cases

100 RILs. HI showed the strongest changes shifting from 0.06 in the top data set to 0.62 when bottom data set was analyzed. Our results indicate that even when improving RUE is a viable strategy to obtain higher yielding soybeans, its relatively low additive-genetic variance might make challenge its enhance using conventional breeding techniques. Methodologies based on genomic information like genomic selection might help to obtain better results for this trait.

Low heritability for HI in the top 100 yielding lines suggests an exhausted contribution of this trait in already highly productive materials leaving room to improve GY through Ei and RUE.

When the whole data set was considered, a significant positive additive-genetic correlation between grain yield and all the three efficiencies were found, with particular strong relationship with HI and RUE (Table 2). In contrast, a negative correlation between radiation use efficiency and fraction of light intercepted was established. Among high yielding lines, grain yield showed a positive additive-genetic correlation with Ei and RUE but null influence of HI. Negative correlation between RUE and HI was determined for this group. When the bottom 100 RILs were considered, positive correlation between RUE and HI with GY was found. Likewise, HI is positively correlated with Ei and RUE. As already mentioned above, these results highlight the importance of

improving Ei and RUE in high yielding materials, considering the reduction in the additive genetic variance for HI. The proposed limited contribution of HI over GY is surely a result of a continuous direct or indirect selection over this trait till reducing its variation.

Association analysis

Regions in the genome potentially associated with Ei, RUE, and HI were identified using a genome-wide association following the whole-genome regression approach (Table 3). The efficiency of light interception was associated with two SNPs on chromosome 7, one SNP on chromosome 11, and another on chromosome 13 (Fig. 2A). Radiation use efficiency showed association with four SNPs located on chromosomes 1, 7, 11, and 18, respectively (Fig. 2B). SNPs found on chromosome 7 for Ei and RUE are not located on the same linkage disequilibrium block. Harvest index, in turn, was associated with two SNPs; the first on the chromosome 13 and the second on chromosome 15 (Fig. 2C). Once again, SNPs located on chromosome 13 for Ei and HI do not share a linkage disequilibrium block. Although grain yield was also analyzed following the same approach, we did not identify specific SNPs potentially associated with the trait at the threshold chosen for this study (Fig. 2D). These findings not only enable the use of more efficient breeding methodologies like marker assisted selection during the breeding process but also encourage us to consider improving physiological traits as a viable strategy to improve GY in soybean.

Discussion

The efficiency of light interception denotes the amount of solar radiation intercepted relative to the total available during the growing season. A linear or at least curvilinear relationship between the cumulative solar radiation intercepted and the total standing dry matter is reported for crops in

Table 2 Additive-genetic correlations for light interception efficiency (Ei), radiation use efficiency (RUE), harvest index (HI), and grain yield (GY) for three data sets: All, Top 100, and Bottom 100 in a maturity-controlled panel of soybean.

Data set	Trait	Ei	RUE	HI	GY
All	Ei	1.00			
	RUE	− 0.12	1.00		
	HI	0.11	0.23	1.00	
	GY	0.20	0.40	0.79	1.00
Top 100	Ei	1.00			
	RUE	− 0.27	1.00		
	HI	0.57	− 0.62	1.00	
	GY	0.41	0.21	− 0.01	1.00
Bottom 100	Ei	1.00			
	RUE	0.20	1.00		
	HI	0.83	0.65	1.00	
	GY	− 0.06	0.59	0.45	1.00

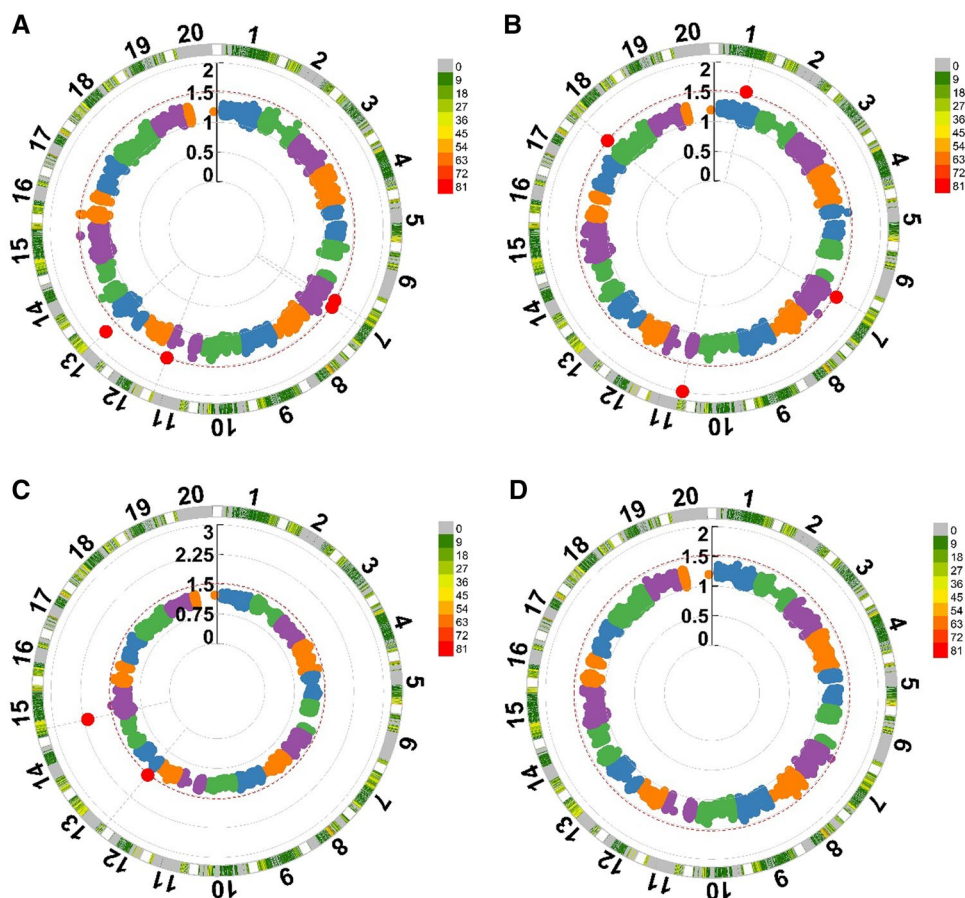
Data set “All” includes three hundred and eight-three recombinant inbred lines RIL. Three environments in all the cases

Table 3 Significant single nucleotide polymorphism (SNP) associated with for light interception efficiency (Ei), radiation use efficiency (RUE), and harvest index (HI) in a maturity-controlled panel of soybean

Trait	SNP	Chromosome	Position	<i>p</i> value	LD Kbp
Ei	13_24980935_T_G	13	24,980,935	0.018	311
	7_22817668_G_A	7	22,817,668	0.029	235
	11_37345229_G_A	11	37,345,229	0.030	176
	7_13739807_C_T	7	13,739,807	0.030	235
RUE	11_6047451_G_T	11	6,047,451	0.010	176
	7_8112122_C_T	7	8,112,122	0.028	235
	1_35840821_A_G	1	35,840,821	0.028	226
	18_9127154_T_C	18	9,127,154	0.030	375
HI	15_7834108_C_T	15	7,834,108	0.007	305
	13_1743515_G_A	13	1,743,515	0.028	311

p value probability of the SNP being included in the model, *LD*—linkage disequilibrium

Fig. 2 Genetic architecture for light interception efficiency **A**, radiation use efficiency **B**, harvest index **C** and grain yield **D** in a maturity-controlled soybean panel. Three hundred and eighty-three recombinant inbred lines (RIL) and three environments. The number outside the circle indicate the chromosome, the red dashed circle represents the threshold expressed as $-\log p$ value, and red dots correspond to the significant SNPs



general (Monteith 1977, 1994; Gosse et al. 1986; Sinclair y Muchow 1999; Ceotto et al. 2013), and soybean in particular (Shibles y Weber 1966; Muchow et al. 1993; Rattalino Edreira et al. 2020). Patterns of light interception in soybean have changed during the last decades; new cultivars show an improved ability to capture light mainly during the reproductive period (Koester et al. 2014) and newer cultivars produce more branches, especially under low density, increasing light interception and seed yield (Suhre et al. 2014). The values of Ei we determined (Fig. 1A) agree with the previously reported range of 40–70% in soybean (Edwards et al. 2005; Koester et al. 2014). Although we did not observe wide range phenotypic variation when evaluated by environment, with a maximum range of ~11%, our results indicate that, on average, 50% of the seasonal solar radiation available is not effectively captured by the canopy (Fig. 1A). Such as large portion of a free resource remaining uncaptured presents an opportunity to capitalize on this trait through agronomic and breeding strategies. Agronomic practices contributing to faster canopy closure are key, considering we observed that at least 60 DAP were needed to achieve full light interception. Plant population, referred to distance between plants and particularly row spacing, is well-documented as an effective approach to increasing Ei (Singer 2001; Edwards

et al. 2005; Wells et al. 2010; Liu et al. 2018) followed by irrigation, plant nutrition, planting date, and soil management (Muchow et al. 1993; Sinclair y Muchow 1999; Li et al. 2008; Sandaña et al. 2012).

Regarding genetic improvement, our results suggest breeding for Ei is feasible because we observed moderate to high h^2 and a strong relationship with GY. Improvements in Ei are key to achieve better GY especially in high yielding materials, whereas their improvement in low-yielding material seems not to be crucial (Table 2) since their canopies are frequently wider and taller to capture light effectively. Augmented light interception especially during the reproductive period is reported as a positive contributor for higher grain yield in new soybean cultivars (Koester et al. 2014). The connection between plant architecture and yield components in soybean was previously established by Zhang et al. (2015) using an association analysis. Their findings of significant association of canopy architecture and yield components agree with the positive significant additive-genetic correlation documented here; however, none of the four SNPs associated with changes in the soybean architecture reported by those authors overlap with the four SNPs we identified. In our results, the SNP with higher $-\log p$ value of association corresponds to a polymorphism located on

chromosome 13 (Fig. 2A) where a previous quantitative trait locus (QTL) for plant height is reported (Li et al. 2010a). Several gene models are also documented for this linkage block covering putative genes, as *Glyma13g21700* a gene associated with the cytochrome P450 (iron ion-binding), *Glyma13g21540* a fructose-bisphosphate aldolase, and heat stress transcriptional factors (Grant et al. 2010). The SNP located on chromosome 7 (Fig. 2A) called 7_13739807_C_T appears in a euchromatic region (Song et al. 2016) where 29 gene models have been proposed including genes associated with photosynthesis (*Glyma07g14227*) and iron binding-cytochrome P450 (Grant et al. 2010). Although the second SNP on the chromosome 7 (7_22817668_G_A) is placed on a heterochromatic region (Song et al. 2016), five gene models were found in this linkage block encompassing nucleic acid binding and phosphate synthase (Grant et al. 2010). The last association is placed on the chromosome 11 (Fig. 2A) in a linkage block rich in genes with up to 51 models reported including mainly proteins linked with carbon metabolism such as sucrose synthase, phosphate decarboxylase, cytochrome, and glutamine synthetase (Grant et al. 2010). No QTLs have been reported for the linkage blocks on chromosomes 7 and 11; thus, our results correspond to the first report for these regions. Previous reports evaluating average canopy coverage, a property influencing light interception, documented association with this trait on the chromosomes 1, 5, 6, 9, 10 and mainly 19 (Xavier et al. 2017). The lack of overlap between the QTLs we observed, and those QTLs previously reported, might be a result of changes in the population conformation and the methodology implemented since we integrated light interception for whole the cycle while Xavier et al. (2017) focused on the first 60 DAP.

Radiation use efficiency or conversion efficiency indicates the amount of dry matter above-ground produced for each unit of solar radiation (total or PAR) effectively intercepted (Sinclair y Muchow 1999). We determined a mean RUE value (Fig. 1B) either equal or slightly higher than those previously reported for soybean (Sinclair y Horie 1989; Muchow et al. 1993; Sinclair y Muchow 1999; Purcell et al. 2002; Singer et al. 2011; Adeboye et al. 2016). Even when our panel included three classes of diversity: HY, HYD, and DA, the values we calculated remain still far from the maximum theoretical of 5.8 g MJ⁻¹ PAR (Long et al. 2006; Melis 2009; Zhu et al. 2010; Fisher et al. 2014a). In the context of future improvement, our results present two types of potential gains in RUE to be considered: the first corresponds to closing the gap between the mean and the maximum calculated estimated to be ~25%, and the second, more hypothetical, since it implies closing the gap between the current mean and the theoretical maximum estimated as ~100% (doubling). Improving RUE might have a positive effect not only in GY as suggested by the positive genetic correlation

found for all the three data sets we analyzed (Table 2), but also in the total biomass production. In this regard, studies involving historical panels point out the rise in the total dry matter as the main factor accounting for changes in GY (Rowntree et al. 2014; Balboa et al. 2018). Populations with initial low-yielding performance might show faster contribution of RUE improvements on GY, but RUE improvements in high yielding material is also expected to have positive results on GY (Table 2).

Nevertheless, achieving important progress in RUE is challenging as the physiological mechanism and genetic control are not fully understood (Reynolds et al. 2000). In this last aspect, this study pioneers the determination of the genetic architecture and contributes with four SNPs associated with this efficiency (Fig. 2B). The SNP with the highest significant association we detected is located on chromosome 11. Based on its location, it is hypothesized this SNP is associated with the putative gene model *Glyma11g08300* encoding for senescence proteins which directly influence the canopy duration (Grant et al. 2010). QTLs for plant height (Sun et al. 2006; Chen et al. 2007), seed weight (Wang et al. 2004; Zhang et al. 2004; Gai et al. 2007; Chen et al. 2007), branching and node number (Chen et al. 2007) are also documented for this region. The SNP on chromosome 7, in turn, is located on a linkage block with 30 putative gene models (Grant et al. 2010) linked with protein kinases, oxidoreductase activity, transmembrane transport, late embryogenesis abundant protein, and metal ion-binding. QTLs for processes influencing biomass production and distribution such as internode length (Josie et al. 2007), leaflet chlorophyll (Li et al. 2010b), and pods per node (Gai et al. 2007) are also reported for this region. Although the SNP on chromosome 1 (Fig. 2B) is placed on a heterochromatic region (Song et al. 2016), 21 genes models mainly associated with iron ion-binding (cytochrome P450), and oxidoreductase activity were found for this linkage block (Grant et al. 2010). Finally, the SNP on the chromosome 18 (Fig. 2B) is located on a linkage block with a total of 36 genes models primary associated with ADP binding, protein transporter, methyl transferase activity, and GTPase activity.

Although a nonlinear relationship between RUE and GY has been previously documented (Sinclair et al. 2004), our results show a moderate linear genetic correlation between these traits (Table 2). This finding aligns with the fact that higher RUE during the filling period is consistently reported as a mechanism for yield improvement in cereals and legumes (Jiang y Egli 1995; Board y Harville 1998; Reynolds et al. 2001; Takai et al. 2006; Foulkes y Reynolds 2014; Zhu et al. 2016). Though our observation of low narrow-sense heritability for RUE might suggest modest gains through traditional breeding (Table 1), the positive additive-genetic correlation supports the hypothesis of breeding for RUE as a viable approach to improve yield potential (Reynolds et al.

2001; Foulkes y Reynolds 2014). This hypothesis is also supported by the trend among classes (Fig. 1) where families classified as HY with high average RUE reported also the highest GY. In addition, the low h^2 points out a potential explanation for the limited progress in RUE improvement during the last decades as reported by Payne et al. (2012), Reynolds et al. (2000), and Luque et al. (2006).

Several strategies to improve RUE and indirectly GY have been attempted or theoretically explored, including biochemical modifications in the enzyme Rubisco which is the most common. Sinclair et al. (2004), for instance, simulated the effect of ~40% increase in Rubisco content in soybean concluding that without parallel nitrogen metabolism improvement this strategy might actually decrease GY. Doubling the specificity of Rubisco was also considered in wheat by Austin (1999) with maximum theoretical increases in photosynthesis not beyond 20%. Other strategies including selecting for higher photosynthesis under light saturation have demonstrated limited influence on RUE, suggesting that improvements at low light intensities might be more appropriate (Reynolds et al. 2000). Although individual biochemical enhancements seem to have limited effect, recently Wu et al. (2019) through a cross-scale model proposed a parallel increase of Rubisco activity, electron transport, and mesophyll conductance as a viable approach to increase RUE and yield in wheat.

Improving the leaf angle (Reynolds et al. 2000), reducing the antenna size (Long et al. 2015), and reengineering the Calvin-Benson cycle (Simkin et al. 2019) have been proposed as effective strategies to improve RUE. Leaf angle targets upright leaves to achieve erectophile canopies with reduced supra-saturation on top and better light distribution throughout all the leaves (Duncan 1971; Duveck 2005). Reducing the number of chlorophyll molecules capturing light energy to feed the photosystems, also called antenna size, would avoid trapping extra energy that saturates the photosystems since it is dissipated as heat and fluorescence rather than electrons transport (Ort et al. 2011). When nitrogen is optimally allocated, changes in antenna size may increase canopy photosynthesis up to 30% (Song et al. 2017a). Finally, although the mechanism is not well understood, manipulating Calvin-Benson cycle particularly through the overexpression of sedoheptulose-1,7-bisphosphatase has shown positive effects in both biomass production and carbon assimilation in Arabidopsis (Simkin et al. 2017), tobacco (Lefebvre et al. 2005; Simkin et al. 2015) and tomato (Ding et al. 2016).

Harvest index is a measurement of reproductive effort since it indicates the proportion of total dry matter above-ground allocated to reproductive or economically important organs (Hay 1995). The mean HI we determined here is either in the range or slightly lower than other reports in soybean (De Costa y Shanmugathasan 2002; Edwards et al.

2005; De Bruin y Pedersen 2009; Adeboye et al. 2016). Differences for the full data set analysis (Fig. 1C) with previous reports are explained by the fact that our population involves recombinant inbred lines rather than commercial cultivar. Although improvement in HI beyond 0.6 seems unlikely (Evans et al. 1980; Foulkes et al. 2011) and undesirable due the risk of lodging (Berry et al. 2007; Foulkes et al. 2011), our results show a scope to rise this trait at least in ~35% to achieve 0.50–0.55 which is the current range reported for maize, wheat, and rice (Sadras y Lawson 2011; Foulkes y Reynolds 2014). Successful cases of yield increase via HI are reported for corn (Luque et al. 2006), wheat (Reynolds et al. 2001), and rice (Zhu et al. 2016), with the last two primarily associated with dwarfing after the green revolution (Evans et al. 1999). In soybean, conclusions about changes in HI and its contribution to final GY during the last decades are mixed. While Balboa et al. (2018) determined the total biomass produced rather than HI as the main factor accounting for increases in GY from 1922 to 2015, Rowntree et al. (2014) report HI along with total dry matter as the primary drivers for better GY in cultivars released from 1923 to 2008. Non-significant changes in soybean HI during the time also agrees with reports from Frederick et al. (1991) and De Bruin and Pedersen (2009), while a positive relationship between HI and year of release found by Kumudini et al. (2001) add more uncertainty to this topic.

When the whole data set was considered, HI is the main efficiency influencing GY according to the additive-genetic correlations (Table 2). Although this trend is followed for low-yielding lines, this trend was not remained when only top yielding RILs were analyzed (Table 2). Lack of variation for HI in high yielding lines is likely associated with its null correlation with GY. This assumption is supported by the low narrow-sense heritability showed by HI in this group (Table 1). This result also reinforces the optimization of Ei and RUE as a priority to achieve higher yielding in future soybean cultivars. Positive correlation found in all the data set and the low-yielding RILs not only agrees with early reports about proportionality between HI and GY (Donald y Hamblin 1976) but also explains why HI has been the recurrent trait directly or indirectly improved through breeding during the last decades (Hay 1995; Luque et al. 2006; Reynolds et al. 2012b; Koester 2014; Fisher et al. 2014b). These changes in HI might have been facilitated by the high narrow-sense heritability on low-yielding lines (Table 1) available some decades ago, which ensured moderate to high genetic gains.

As it was already mentioned, reports in other crops highlight reductions in height as a major driver for increased HI; however, enlarged sink sizes via a greater number of spikelets per panicle and photosynthesis in the flag leaf are also a key contributor in rice and wheat (Zhu et al. 2016; Carmo-Silva et al. 2017). Recently, a positive significant genetic

correlation between photosynthesis during the early filling period ($R5$) and grain yield in soybean was demonstrated (Lopez et al. 2019). While changes in HI during the time in soybean are unclear, studies involving historical panels suggest shifts in parameters linked to the sink-source relationship such as the number of pods in branches (Suhre et al. 2014) and extended seed-filling and reproductive period (Rowntree et al. 2014). Altering sink-source relationship in soybean to favor sink strength appears as an alternative to impact HI and GY considering new cultivars under no stress conditions seem to be sink limited (Egli y Bruening 2001; Liu et al. 2006; Ainsworth et al. 2012). Parallel to sink strength changes, improvements in the long-distance transport particularly during the pre-storage phase ($R3$ – $R6$) are also proposed to move the surplus of carbohydrates into new sources (Borrás et al. 2004; Ainsworth et al. 2012).

Closing the gap between the current and the maximum attainable HI seems feasibly only if this efficiency is actively considered into the breeding processes. To start this integration, this study presents the genetic architecture for harvest index (Fig. 2C). The GWA analysis revealed two SNPs associated with this trait. The first SNP corresponds to the polymorphism with highest $-\log p$ value and is placed on the chromosome 15 in a region where up to 51 gene models involving primarily with energy production such as protein kinases and ATP binding were documented (Grant et al. 2010). The second SNP, in turn, is located on a region where QTLs for seed abortion, seed set, seed weight, and plant height have been already reported (Tischner et al. 2003; Kato et al. 2014; Yao et al. 2015). Likewise, in this region, up to 29 gene models are documented encompassing nucleic acid, RNA, and protein binding, hydrolase activity, transmembrane transport, and electron carrier activity (Grant et al. 2010).

Conclusions

Improvement in grain yield via breeding for Ei, RUE, and HI is feasible considering the phenotypic variation, genetic correlation and heritability found in this study. Enhancements in RUE and Ei must be considered long-term goals since both the gap between the current mean and the maximum documented here and the current mean and the theoretical maximum remain quite large. Working on optimizing the HI to close the gap between current and maximum achievable values is useful, but it is possible that the contribution of this optimized parameter is not as high as expected particularly when breeding using high yielding germplasm. Low yield in soybean is genetically associated with reduced capability to transform solar energy into biomass and poor reproductive effort rather than limited ability to capture solar radiation.

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Author contribution statement MAL and KMR conceived and designed the experiments. MAL and FFM collected the field data. KC oversaw the collection of UAS imagery and ground reference data. AH completed the image analyses. MAL conducted the data analysis and interpretation. MAL wrote and edited the manuscript. KMR coordinated-supervised the research and ensured funding.

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Data availability Information about the parents, complete description of crosses, and genomic information is available in <https://www.soybase.org/SoyNAM/> and Supplementary Table 1 and 2. Additional information is available under request.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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